

Alexander Nickita, Embedded Software Engineering | Mathematics | Researcher @ Lawrence Technological University

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LINKS

LinkedIn, GitHub

PROFILE

Embedded Software Engineering student at Lawrence Technological University with a Mathematics minor, specializing in low-level programming, real-time systems, and microcontroller/microprocessor development. Skilled in C/C++, Java, and logic design with a foundation in circuit analysis and data structures. Experienced in applying engineering principles to embedded platforms, with strong problem-solving skills and familiarity with Agile practices, UML modeling, and testing methodologies.

EMPLOYMENT HISTORY

May 2026 — Aug 2026	Embedded Software Engineer (Intern), General Dynamics Land Systems Sterling Heights, Michigan Developing embedded software for advanced ground vehicle systems in C/C++ within a defense-grade bare-metal environment. Collaborating with a cross-functional engineering team to support real-time system requirements, hardware-software integration, and firmware validation at the subsystem level.
May 2025 — Present	Research Assistant, Lawrence Technological University Southfield, Michigan, United States Help principal investigators with experimental design, variable definitions & scope, as well as help to proctor experiments and find subjects. I also mentor and oversee other undergraduate students who participate in research at LTU
Nov 2024 — Present	Computer Assistant, Southfield Public Library Southfield, Michigan, United States

EDUCATION

Jan 2027 — Present	Bachelor of Applied Science - BASc in Embedded Software Engineering, Lawrence Technological University Expected Graduation: May 2027 GPA: 3.60
Jan 2020 — Jan 2023	Associate's degree in Computer Software Engineering, Oakland Community College Graduated with honors, 3.75 GPA, Magna Cum Laude

ACADEMIC PROJECTS

	Bare-Metal Smart Home Controller (TM4C123G) Lawrence Technological University Designed and implemented a bare-metal IoT smart home controller on the TM4C123G microcontroller using C. Integrated Bluetooth (HC-05), DHT11 temperature/humidity sensing, WS2812B addressable LED control, PIR motion detection, and fan speed control via PWM. Developed custom UART, SPI, and GPIO drivers from scratch with no OS or HAL, producing full technical documentation and demo materials.
	Edge Vision Node — Embedded Smart Home Sensing Module Lawrence Technological University Developing a low-cost embedded IoT sensing module on Raspberry Pi 5 with a camera module, environmental sensors (BME280, PIR), and ESP32 nodes. Uses OpenCV and TensorFlow Lite for on-

device real-time human presence detection and activity classification, communicating with a smart home control system via standard IoT protocols to trigger automated actions without cloud dependency.

Computer Vision-based Identification Application

Lawrence Technological University

Using OpenCV, trained an AI model to be able to distinguish humans from other objects, powered by Tiva C-series embedded microcontroller

SKILLS AND CERTIFICATIONS

C/C++	Digital Logic Design
Python	Circuit Analysis
Bare-Metal Firmware	Microcontroller Programming
UART / SPI / I2C	Sensor Integration (DHT11, PIR, WS2812B)
SAFe 6 Scrum Master	Bluetooth / BLE (HC-05)
Data Structures	OpenCV / TensorFlow Lite
	Raspberry Pi / Linux
	IoT Protocols (MQTT, BLE)

AWARDS

Dean’s List Fall 2024	Lawrence Technological University
Dean’s List Spring 2025	Lawrence Technological University
Dean’s List Fall 2025	Lawrence Technological University
Dean’s List Spring 2026	Lawrence Technological University
Outstanding Member of a Student Organization (IEEE)	Lawrence Technological University
Virinder K. Moudgil Merit Scholarship	Lawrence Technological University
Magna Cum Laude	Oakland Community College
Phi Theta Kappa Honors Society member	Oakland Community College